WORKED SOLUTION

STEP II 2010, Question 1

Pure • Curvature and Osculating Circles

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q1

The key idea

"Touches" means the circle and the curve share the point P and the gradient there; "the rate of change of the gradient" means $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$. So we need a circle whose y, $\frac{\mathrm{d}y}{\mathrm{d}x}$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ all agree with the curve's at P. Differentiating the equation of a general circle implicitly, twice, turns this into simple equations for the centre.

Solution

The curve at P

For $y = 1 - x + \tan x$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -1 + \sec^2 x = \tan^2 x, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2\tan x \sec^2 x.$$

At $x = \frac{1}{4}\pi$ we have $\tan x = 1$ and $\sec^2 x = 2$, so

$$y = 2 - \tfrac{1}{4}\pi, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = 1, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4.$$

So $P = (\frac{1}{4}\pi,\ 2 - \frac{1}{4}\pi)$.

The circle

Let the circle be $(x-a)^2 + (y-b)^2 = r^2$. Differentiating implicitly with respect to x,

$$(x-a) + (y-b)\frac{\mathrm{d}y}{\mathrm{d}x} = 0,$$

and differentiating again (using the product rule on the second term),

$$1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + (y-b)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0.$$

Ask yourself Why can we put the curve's values straight into the circle's equations?

These two equations hold at every point of the circle, with y, $\mathrm{d}y/\mathrm{d}x$ and $\mathrm{d}^2y/\mathrm{d}x^2$ being the circle's own values. The two conditions say that, at P, the circle passes through P with the same gradient as C and the same second derivative. So at P we may use $x = \frac{1}{4}\pi$, $y = 2 - \frac{1}{4}\pi$, $\mathrm{d}y/\mathrm{d}x = 1$ and $\mathrm{d}^2y/\mathrm{d}x^2 = 4$.

The second equation gives

$$1 + 1 + 4(y - b) = 0 \quad \Longrightarrow \quad y - b = -\tfrac{1}{2} \quad \Longrightarrow \quad b = 2 - \tfrac{1}{4}\pi + \tfrac{1}{2} = \tfrac{5}{2} - \tfrac{1}{4}\pi.$$

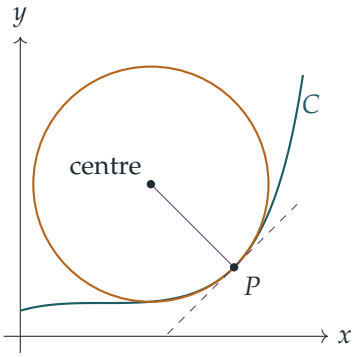
The first then gives

$$x - a = -(y - b) \cdot 1 = \tfrac{1}{2} \quad \Longrightarrow \quad a = \tfrac{1}{4}\pi - \tfrac{1}{2}.$$

Finally, $r^2 = (x - a)^2 + (y - b)^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. So the osculating circle has

$$\text{centre } \left(\tfrac{1}{4}\pi - \tfrac{1}{2},\ \tfrac{5}{2} - \tfrac{1}{4}\pi\right) \qquad \text{and radius} \qquad \frac{1}{\sqrt{2}}.$$

It makes sense geometrically. The centre lies a distance $\frac{1}{\sqrt{2}}$ from P along the direction $(-1, 1)$, which is perpendicular to the tangent of gradient 1. It lies on the side where $y > 2 - \frac{1}{4}\pi$, that is, above the curve, which is right because $\mathrm{d}^2y/\mathrm{d}x^2 > 0$ and the curve is concave upwards there.



Checks

- The general radius-of-curvature formula $\rho = \left(1 + y'^2\right)^{3/2} / |y''|$ gives $2^{3/2}/4 = 1/\sqrt{2}$, as found.
- Numerically, the centre is about $(0.2854,\ 1.7146)$ and P is about $(0.7854,\ 1.2146)$; their distance is 0.70711. Taking the lower half of the circle, $y = b - \sqrt{r^2 - (x - a)^2}$, and differentiating numerically at $x = 0.7854$ gives gradient 1.0000 and second derivative 4.000, matching the curve.

Common slips

- Placing the centre on the wrong side of the tangent: the sign of $y - b$ is fixed by the second equation, so let the algebra decide.
- Forgetting that differentiating $(y - b)\,\mathrm{d}y/\mathrm{d}x$ needs the product rule, which is where the $(\mathrm{d}y/\mathrm{d}x)^2$ term comes from.

Beyond STEP

The radius of the osculating circle is the radius of curvature, and its reciprocal $\kappa = |y''|/(1 + y'^2)^{3/2}$ is the curvature, the central quantity in the differential geometry of curves. For a curve in space, curvature joins torsion in the Frenet–Serret formulae, which describe how a curve bends and twists. Siklos, *Advanced Problems in Mathematics*, has more STEP questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.